Effects of Alcohol on the Liver

Aim: To observe the changes caused to body tissue by alcohol.

Equipment:

- 3 pieces of animal liver
- Ethanol
- 10cm^3 measuring cylinder
- Top pan balance
- Weighing boat
- Forceps or gloves
- 3 small beakers
- Ruler
- Plastic food wrap
- Camera (optional)

Safety Information



Liver – possible bacterial presence

Handle using forceps or gloves and wash hands after handling

Glassware – risk of cuts if broken

Apply pressure to wounds and inform teacher immediately

Method:

Step 1: Measure 20ml of ethanol into each of the three beakers.

Step 2: Use the top pan balance to accurately measure the starting mass of each piece of liver. Don't forget to calibrate the balance to account for the mass of the weighing boat. Record your observations.

Step 3: Using a ruler wrapped in food wrap, measure the width and length of each piece of liver. Record your observations.

You might want to take a photograph of the liver before and then again after steps 5-7, for comparison.

Step 4: Place a piece of liver into each of the beakers and cover with plastic food wrap. Set to one side.

Step 5: Examine the first beaker after 24 hours. Measure the mass, width and length of the tissue. Record your observations.

Step 6: Examine the second beaker after 3 days. Measure the mass, width and length of the tissue. Record your observations.

Step 7: Examine the third beaker after 1 week. Measure the mass, width and length of the tissue. Record your observations.

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Expanding the investigation

- You might plan to investigate how different types of alcohol affect the tissue (beers, spirits, wines, alcopops etc).
- You might plan to investigate how the alcoholic content (%) affects the tissue.



Your liver is a large, wedge-shaped, spongy organ. It is located under the diaphragm, on the right hand side. The liver has many functions in your body including the removal of toxins from the body, the regulation of blood sugar and the production of bile for digestion.

Results table:

	Beaker 1	Beaker 2	Beaker 3
Time left	24 hours	3 days	1 week
Observations before (colour, texture)			
Observations after (colour, texture)			
Starting mass			
End mass			
Starting width			
End width			
Starting length			
End length			

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Things to think about

- How has the liver tissue changed in the presence of ethanol?
- How do you think this will affect the function of the liver?
- How might the effects of the liver function affect the person as a whole?
- What advice or recommendations could you make based on the evidence you collected?

Extra Challenge

Using your findings and your own knowledge, produce a small message that could be displayed on alcohol packaging (like the ones you see on tobacco products).

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